

H1 — EEoS haul-level biomass prediction: results summary

NS-IBTS Q1 · 1985–2015 · 12,069 hauls

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Contents

Figure 1 — Headline result	1
1. Key statistics	2
1.1 $\log_r2 = -0.223$ — predictive R^2	2
1.2 $\text{cor}(\log B)^2 = 0.926$ — rank agreement (diagnostic only)	2
1.3 $\ln(E) \log_r2 = 0.595$ — correlative benchmark	3
2. How B_obs and B_pred are compared	3
3. Null model: $p < 0.001$	3
4. Are the inputs driving the signal, or the theory?	4

Figure 1 — Headline result

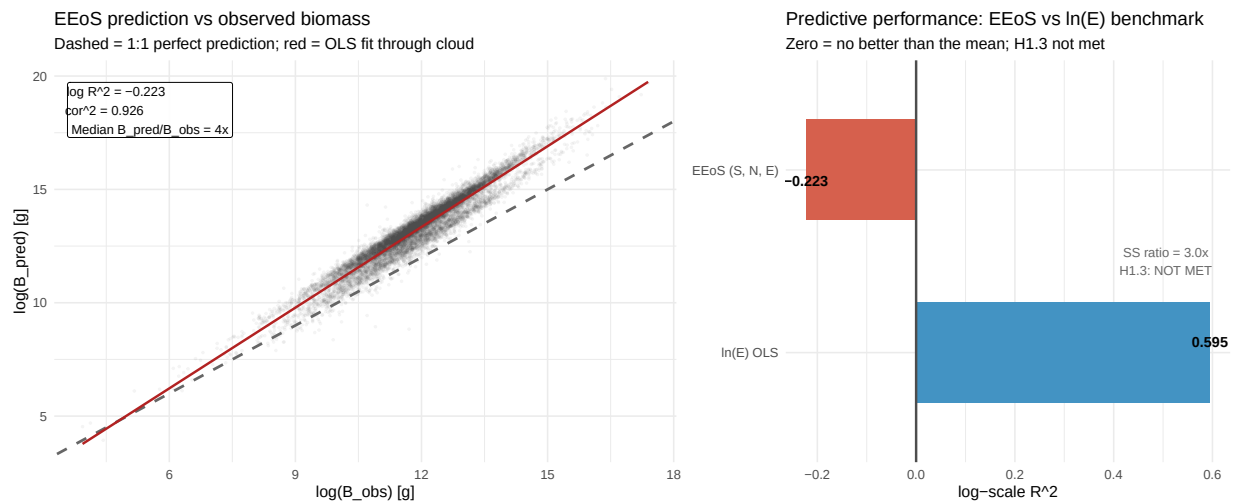


Figure 1. *Left:* $\log(B_obs)$ vs $\log(B_pred)$ for all hauls. The dashed line is the 1:1 perfect-prediction line, where prediction equals observation exactly. The cloud sits systematically above it throughout the biomass range, indicating consistent overprediction (median $B_pred/B_obs = 4\times$). B_pred/B_obs are both in grams. Rank agreement is high (Pearson $r^2 = 0.93$) but the magnitude is wrong. *Right:* log-scale R^2 for EEoS versus a fitted OLS regression of $\log(B_obs)$ on $\log(E)$ alone. EEoS (red, -0.22) does not clear

zero; $\ln(E)$ OLS (blue, +0.60) does. The SS ratio of $3.0\times$ means EEoS leaves three times more unexplained variance than metabolic rate with a fitted line.

EEoS does not predict absolute haul-level community biomass: $\log_r2 = -0.223$, below both the zero bar (predicting the mean) and performs worse than a simple metabolic-rate regression. A strong rank correlation ($\text{cor}^2 = 0.926$) reflects that the inputs covary with biomass, not that the theory predicts it correctly.

1. Key statistics

1.1 $\log_r2 = -0.223$ — predictive R^2

What it measures

$\log_r2 = 1 - \text{SS_res} / \text{SS_tot}$, computed on log-transformed biomass values.

- SS_res : summed squared differences between $\log(\text{B_obs})$ and $\log(\text{B_pred})$ = unexplained variance after using EEoS.
- SS_tot : total variance in $\log(\text{B_obs})$ = how much hauls differ from one another in observed biomass.
- Score of +1.0: EEoS accounts for all variation.
- Score of 0.0: EEoS does no better than predicting the mean $\log(\text{B_obs})$ for every haul.
- Score of -0.223: predictions add noise rather than reducing uncertainty.

No parameters are adjusted to fit the data. EEoS outputs a specific prediction for each haul from (S, N, E) with no free parameters, evaluated exactly as given.

Why it is the right metric for H1

H1 is a prediction test, not a correlation test. $\log_r2$ is the only metric that penalises getting the absolute scale wrong. A theory that predicts every haul at $4\times$ the true value would score a strongly negative $\log_r2$ and a near-perfect correlation — these are different claims, and conflating them would constitute a false positive for H1.

Note: raw-scale $R^2 = -91.216$, more negative still. Log scale is appropriate because biomass spans four orders of magnitude across hauls; on the raw scale, R^2 is dominated by a handful of the largest hauls.

1.2 $\text{cor}(\log \text{ B})^2 = 0.926$ — rank agreement (diagnostic only)

What it measures

Squared Pearson correlation between $\log(\text{B_obs})$ and $\log(\text{B_pred})$. Whether hauls rank in the same order under observed and predicted biomass — regardless of whether the magnitudes match.

Why it is reported as a diagnostic, not the H1 test

EEoS consistently overpredicts catch biomass by approximately $4.1\times$. cor^2 does not capture this. The 0.93 value is largely the E signal: metabolic rate covaries with catch because more fish caught means higher total metabolic activity. A model that doubles every prediction would have the same cor^2 . If cor^2 were the H1 test, the systematic overprediction would be invisible in the result.

The two numbers together explain the main scatter: tight and well-organised ($\text{cor}^2 = 0.926$, rank agreement) but sitting consistently above the 1:1 line ($\log_r2 = -0.223$, scale wrong).

1.3 $\ln(E)$ $\log_r2 = 0.595$ — correlative benchmark

What it measures

$\log_r2$ from an OLS regression of $\log(B_obs)$ on $\log(E)$ alone — a fitted straight line using only metabolic rate, with no equation of state:

$$\log(B_obs) = 4.32 + 0.714 \times \log(E)$$

Fitted on the same 12,069 hauls. Because OLS optimises its slope and intercept to minimise SS_res , this R^2 cannot go below zero.

Why it matters

Following Harte et al. (2022), who proposed $\ln(E)$ as the natural reference model, EEoS is expected to outperform a simple metabolic-rate regression.

The result — $SS\ ratio = 3.02\times$ — means EEoS leaves 3.02 times more unexplained variance than a fitted E regression. E alone achieves $\log_r2 = 0.595$ with a fitted line; EEoS using all three state variables achieves -0.223. The additional structure the equation of state imposes is not improving prediction at this scale.

2. How B_obs and B_pred are compared

Both are in grams. B_obs is the sum of individual fish weights per haul from FishGlob. B_pred requires a conversion:

$$B_pred\ (grams) = B_pred_norm \times m_min$$

EEoS outputs a dimensionless quantity (B_pred_norm) because the theory works in units of the minimum individual metabolic rate. m_min is the mass of the smallest individual in the haul, derived from the smallest length bin using the same FishBase length-weight parameters that build S, N, and E. The log-scale residual $\log(B_obs) - \log(B_pred)$ is equivalent to $\log(B_obs/B_pred)$ — a dimensionless measure of over- or under-prediction.

3. Null model: $p < 0.001$

What it tests

Whether the haul-specific pairing of $(S, N, E) \rightarrow B_pred$ with that haul's actual B_obs carries more information than random biomass assignment.

B_obs is randomised while B_pred is held fixed from real state variables. $\log_r2$ is recomputed across 999 permutations, providing a distribution of “what $\log_r2$ looks like with no signal.” The one-sided p-value is the fraction of null $\log_r2$ values that meet or exceed the observed.

S, N, and E cannot be shuffled — EEoS requires $E > N$ and integer state variables, constraints that permutation routinely violates.

Why the uniform null, not a simple B shuffle

$\log(B_{\text{obs}})$ is strongly peaked: its histogram peak is approximately $5\times$ higher than a uniform distribution across the same range, and the IQR covers only 12% of the full range. A B shuffle resamples mostly from the same narrow central mass, producing null predictions close to the observed distribution and a lenient test. The primary null draws B_{obs} uniformly from the central 95% of the $\log(B_{\text{obs}})$ range, spreading randomised values broadly and setting a harder baseline. The B shuffle is retained as a robustness check.

Result

Table 1: Null permutation results under two schemes.

Null scheme	Null median $\log R^2$	Observed $\log R^2$	p (one-sided)
Uniform on 95% $\log(B_{\text{obs}})$	-2.02	-0.223	< 0.001
B shuffle	-2.60	-0.223	< 0.001

The observed $\log_{\text{r}2}$ of -0.223 lies far above both null medians. The haul-specific state variables carry genuine information about that haul’s biomass. Beating the null does not mean EEoS predicts accurately — $\log_{\text{r}2}$ remains negative. The null establishes the signal is real; $\log_{\text{r}2}$ establishes it is too weak for useful absolute prediction.

4. Are the inputs driving the signal, or the theory?

S, N, and E are not independent of B_{obs} . Hauls that catch more fish have higher N, higher total metabolic activity (E), and likely higher S. Any function of those variables will tend to correlate with B_{obs} . The cor^2 of 0.926 largely reflects this: E alone achieves $\log_{\text{r}2} = 0.595$ with a fitted line; EEoS using the full partition function achieves -0.223. The additional structure the equation of state imposes relative to a simple E regression is negative.

The residual distribution is unimodal, with a single peak at approximately -1.5 log-units and a right-skewed tail toward zero (Supplementary Figure S1).

The median $4.1\times$ overprediction is not uniform across the biomass range.

Table 2: EEoS overprediction ratio by biomass quartile.

Quartile	B range ($\log g$)	Median ratio	$\log_{\text{r}2}$
Q1 (smallest)	3.9–11.1	3.1x	-1.12
Q2	11.1–11.9	3.9x	-32.77
Q3	11.9–12.7	4.5x	-40.13
Q4 (largest)	12.7–17.4	5.5x	-5.34

The overprediction ratio increases monotonically with haul biomass: the largest hauls are overpredicted by approximately $5.5\times$, the smallest by approximately $3.1\times$. A fixed catchability correction — the simplest explanation for a systematic offset — would produce the same ratio across all panels. The scale-dependence means high-biomass assemblages are disproportionately over-predicted by the theory.

The $\log_{\text{r}2}$ values in Q2 and Q3 are statistical artifacts. Each middle quartile spans less than 0.8 log-units of B_{obs} ; within that narrow x-range, EEoS predictions still vary considerably (driven by E, S, and

N independently), so SS_{res} is large relative to the small SS_{tot} for that slice. The median ratio is the reliable metric within each quartile.

The scale-dependent offset is relevant for interpreting H2 and H3. If spatially structured fishing pressure systematically shifts hauls toward lower-biomass quartiles in high-pressure areas, the offset gradient itself may generate spatial structure in EEoS residuals independently of any disturbance signal. This needs to be accounted for in downstream attribution.

Conclusion to H1:

At haul scale in a trawl survey, EEoS does not correctly translate the information in S, N, and E into absolute biomass. Whether EEoS residuals carry structured information about disturbance — which does not require accurate absolute prediction — is the question H2 and H3 address.

*Numbers from `outputs/h1_model_comparison.rds`, `outputs/h1_lne_coefficients.csv`, and `outputs/null_distribution`.
Supplementary figures: `h1_two_trends_investigation.html`. Full diagnostic plots: `review_supervisor_h1_results_details`*